

APPLICATION PROGRAMMING INTERFACES (API)

API = Application Programming Interface

Interface = allows interactions with the system

- Ex : buttons on a drink dispenser
- Ex : mouse, keyboard, computer screen
- Ex : software graphical interface (windows, buttons, drop-down lists, and so on)

API allows interactions between a software and other softwares.

- Ex : The **mobile application** of your bank will interact with the **database** of your bank
- **Word** interacts with **Office 365** to save your files online

API is the protocol each software must follow to communicate

Protocol means available functions/rules to interact :

- Ex : Retrieving the current account balance : parameter account number
- Ex : Saving a file in the Cloud : content parameter and file name

API is an interface allowing interactions with the software (database, cloud, and so on)

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API REST : Representational State Transfer

API REST use HTTP (like websites) : means of communication

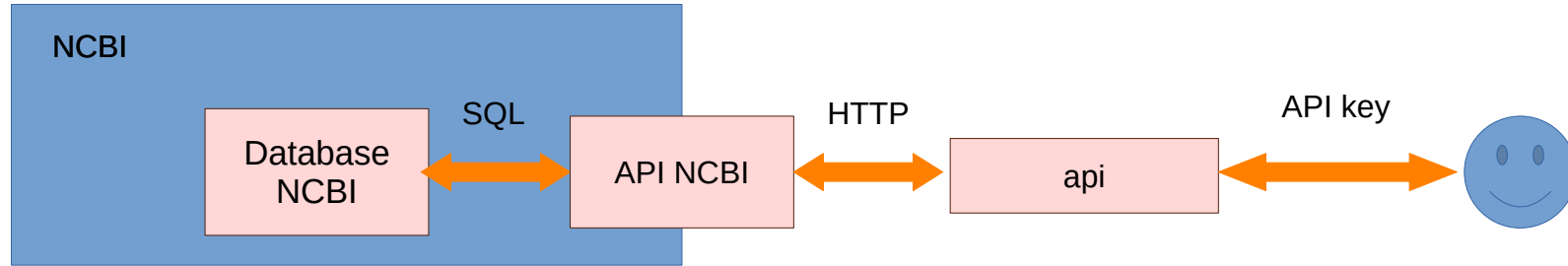
The way you interact with an API is identical to the way you interact with a website

A HTTP request is composed of a URI, a method and one or several header(s)

- HTTP uses methods :
 - GET : retrieve/read
 - POST : create
 - PUT : modify
 - DELETE : delete
 - INFO : metadata (information about data, different to data)
- HTTP uses URI/URL (unique resource identifier/link)
 - A URI/URL identifies a unique resource
 - We can use different methods on a same resource (POST, GET, PUT, DELETE)
- Headers give complementary information : ex : API key

URI/URL states what we are working on, the methods states what we are doing with it

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API key : Identifies the software (api) in connection with the API (NCBI API)

=> Why ?

- Limiting call frequency : ex : max 3 requests / minute
- Identifying the softwares which use this API

Additional authentication mechanisms might be necessary to identify the user (ex : banks)

An API key is a private identifier

=> It is as sensitive as a password

=> We must not hard-code the API key into the code, nor commit it to the repository

Good practices : environment variables (bashrc), reading hidden files (no commit)